# DEVELOPMENT OF EDUCATION IN THE WORLD: A SUMMARY STATISTICAL REVIEW

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*Using data from 1960 to 1976, the author examines population changes, especially at the school-age level. He then shows how the enrolment increase – 77% for the period – is distributed at successive levels of education and between developed and developing countries. The comparison of enrolments with total populations produces a series of enrolment ratios. Despite the considerable progress recorded by developing countries, the date show that drop-out rates remain high.*

*Public expenditures on education show an even greater rise than might be expected from enrolments. Of significance here is the growing percentage of GNP devoted to education, particularly in the developing countries.*

*Aufgrund von Daten aus der Zeit von 1960 bis 1976 untersucht der Verfasser die Veränderungen in der Bevölkerung, speziell auf der Altersstufe der Schulpflichtigen. Dann zeigt er, wie sich die Zunahme an Schulkindern – 77% während der Berichtszeit – auf die aufeinanderfolgenden Bildungsstufen sowie auf die Industrie- und Entwicklungsländer verteilt. Aus dem Vergleich zwischen der Anzahl an Schulkindern und der jeweiligen Gesamtbevölkerung ergeben sich unterschiedliche Verhältniszahlen. Zwar berichten die Entwicklungsländer von beachtlichem Fortschritt, aber die Daten zeigen, daß die Anzahl der vorzeitig die Schule Verlassenden hoch bleibt.*

*Die Ausgaben der öffentlichen Hand für Bildung sind sogar noch höher angestiegen, als nach der Anzahl der Schulkinder zu erwarten wäre. Bedeutsam ist in diesem Zusammenhang der wachsende Anteil des Nationalproduktes, der auf die Bildung verwendet wird, insbesondere in den Entwicklungsländern.*

*Utilisant des data de 1960 à 1976, l'auteur examine les changements dans la population, principalement au niveau de l'âge scolaire. Il montre ensuite comment l'accroissement de la fréquentation scolaire – 77% pour la période en question – est réparti selon les niveaux successifs d'éducation et entre les pays développés et en voie de développement.*

*La comparaison entre la fréquentation scolaire et la totalité de la population donne une série de taux de fréquentation. En dépit du progrès considérable enregistré par les pays en voie de développement, les data révèlent que le taux des abandons en cours d'études demeure élevé.*

*La dépense publique relative à l'éducation montre un accroîssement supérieur à celui que l'on aurait pu attendre d'après la fréquentation scolaire. Le pourcentage croissant du P.N.B. (Produit National Brut) consacré à l'éducation, surtout dans les pays en voie de développement, est particulièrement significatif.*

*International Review of Education – Internationale Zeitschrift für Erziehungswissenschaft – Revue internationale de pédagogie XXV (1979), 147-166.* *All rights reserved.*
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*Introduction*

The quantitative development of education in the last 20-25 years has been an element common to all countries in the world, the less developed as well as the more developed. Education is nowadays universally recognized as a major factor affecting economic and social development and its numerical expansion has become one of the main concerns of present day policy-makers.

However, the tremendous enrolment growth which has taken place since the Second World War has not been sufficient to prevent the existence of large disparities in the provision of education, not only between developed and developing countries but also between and within the latter countries. Furthermore, the problem of growing disparities allied to inefficiency, low quality, increasing costs and unemployment among school leavers has recently put the school system in question though often without suggesting suitable alternatives.

Whatever the value of the above criticisms, it is nonetheless true that many countries have been experiencing quite different patterns of enrolment increases determined, by definition, not only by the relative size of their school-age populations but also by social, economic and political factors, raising sometimes major issues due either to the resulting financial burden or similar constraints.

The following summary review is intended to highlight the salient quantitative patterns of the development of education during the period 1960 until 1976, i.e. the period for which comparable and comprehensive data are presently available in the Unesco Office of Statistics.

To simplify the presentation, world features (excluding China and the Democratic People's Republic of Korea, for which time series are not available) will be grouped into two major areas – developed and developing countries. Comments and data will further be presented for Northern America, and the USSR as major regions in the former areas and Africa, Asia and Latin America for the latter. (Japan is included in Asia as well as in the group of developed countries). We will (1) review the trends and prospects of the school-age population, and (2) outline trends in enrolment, teaching staff and expenditure.

*Population*

The demand for education is heavily influenced by the size and rate of

growth of the school age population. Two sets of indices will help to grasp the different nature of the problem for the developed and the developing countries as presented below:

| | Average Annual rate of population growth | | Percentage of total population aged 5-14 years | |
|---|---|---|---|---|
| | 1960-70 | 1970-75 | 1960 | 1975 |
| Developing countries | 2.28 | 2.36 | 25 | 25 |
| Developed countries | 1.05 | 0.88 | 19 | 17 |

In addition to a higher population growth rate (the extreme values around 1975 were Northern Europe, 0.4%, and Central America, 3.2%), the younger age structure of the population in developing countries (in relative terms their school age population is nearly 50% larger than in the developed countries) places a very heavy burden on their population of working age. This can easily be illustrated by the absolute growth of the age-group 6-11 years old which for most countries is the population of primary school age. Table 1 below indicates that the number of children in this age-group increased by 115 million between 1960 and 1976. They all belong to the developing countries. By region, the development was: 74 million in Asia, 22 million in Africa and 19 million in Latin America. The size of the group remained practically unchanged in the developed countries during the period.

Table 1. *Population aged 6-11 years (in millions)*

| | 1960 | 1976 | Absolute increase |
|---|---|---|---|
| WORLD* | 318 | 433 | 115 |
| Developed countries | 111 | 111 | 0 |
| Developing countries | 207 | 322 | 115 |
| Africa | 44 | 66 | 22 |
| Asia* | 143 | 217 | 74 |
| Latin America | 34 | 53 | 19 |

* Excluding China and the Democratic People's Republic of Korea.

The U.N. Population projections until 2000 allow an examination of the implications of the decline of mortality and changes in fertility as regards the size of school-age population. Thus, taking as an example the age-group 5-14 it is interesting to note that in spite of a fall in the rate of growth for the developing countries as a whole, this group would, between 1960 and 2000, double in Asia, increase 2.7 times in Latin America and triple in Africa. For the developed countries, on the other hand, while the total population is expected to increase (Northern America, 50%; Europe and the USSR, 33%) the growth rate of the population of the 5-14 age-group was decreasing between 1960 and 1975 and is expected to increase between 1975 and the year 2000 as shown in figure 1 below. Thus, whereas developing countries will be faced with an increasing pressure for more places to enrol their growing number of children, developed countries must take account of the consequences of the increases and decreases in school-age population. These differences in future prospects for the population of primary school age have obvious implications for the future requirement for teachers and financial resources.

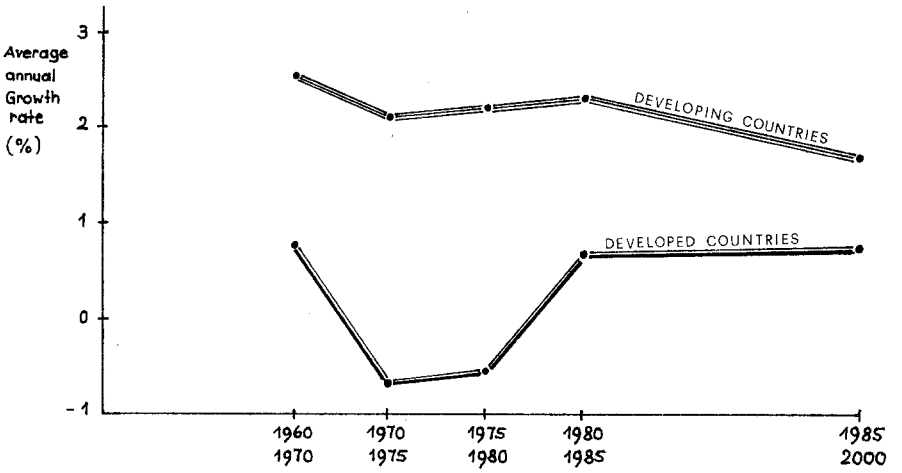



Figure 1. *Growth rate of 5-14 age-group, 1960-2000*

*Enrolment Trends: 1960-1976*

1. *Enrolment by level of education.* In the sixteen-year period between 1960 and 1976, the total enrolment in first, second and third levels of education rose from 328 to 581 million, which means an absolute increase of some

253 million or 77% (i.e. an average annual growth rate of 3.6%). The data are shown in Table 2 and we note the following observations.

(i) The increase was most important at the first level education (136 million) followed respectively by second level (89 million), and third level (28 million).

(ii) More than three-quarters of the absolute increase was accounted for by the developing countries (194 million as compared to 59 million for the developed countries). The former multiplied their 1960 total enrolment by 2.33 and the latter by 1.32.

Table 2. *Development of Enrolment by Level in the World and Major Areas,* 1960-1976 (in millions)*

| Level | Enrolment | | | | Increase 1960-76 | |
|---|---|---|---|---|---|---|
| *The World** | *1960* | *1965* | *1970* | *1976* | *Absolute numbers* | *multiplier* |
| 1st level | 245 | 299 | 343 | 381 | 136 | 1.55 |
| 2nd level | 70 | 100 | 126 | 159 | 89 | 2.27 |
| 3rd level | 13 | 20 | 28 | 41 | 28 | 3.15 |
| Total | 328 | 419 | 497 | 581 | 253 | 1.77 |
| *Developed countries* | | | | | | |
| 1st level | 125 | 135 | 140 | 130 | 5 | 1.04 |
| 2nd level | 47 | 62 | 70 | 81 | 34 | 1.72 |
| 3rd level | 10 | 15 | 21 | 30 | 20 | 3.00 |
| | 182 | 212 | 231 | 241 | 59 | 1.32 |
| *Developing countries* | | | | | | |
| 1st level | 120 | 164 | 203 | 251 | 131 | 2.09 |
| 2nd level | 23 | 38 | 56 | 78 | 55 | 3.39 |
| 3rd level | 3 | 5 | 7 | 11 | 8 | 3.67 |
| | 146 | 207 | 266 | 340 | 194 | 2.33 |

* See footnote at the end of Table 1.

(iii) The distribution of the increase by level varies considerably between the developed and developing countries. In the developed countries, 58% of the increase was at second level, one-third at third level and only 8.5% at primary level. It should of course be noted that education was

already universal at the beginning of the period for primary education and that the corresponding school-age population decreased in absolute numbers during the period. On the other hand, two-thirds of the increase in the developing countries was registered in first level education, 28% in second level and the remaining 4% in higher education.

(iv) As a result of the above changes, the share of the developing countries in the world primary school enrolment increased from less than one-half in 1960 to nearly two-thirds in 1976. Similarly, their share of total enrolment in second level (one-third in 1960) increased to 49%. The share in higher education (one-quarter in 1960), remained almost stable throughout the period.

(v) The educational pyramid consequently changed as follows, taking absolute enrolment in each level, every year, as a ratio to first level enrolment (presented as 100):

| | | *1960* | *1965* | *1970* | *1976* |
|---|---|---|---|---|---|
| Developed countries | 1st | 100 | 100 | 100 | 100 |
| | 2nd | 38 | 46 | 50 | 62 |
| | 3rd | 8 | 11 | 15 | 23 |
| Developing countries | 1st | 100 | 100 | 100 | 100 |
| | 2nd | 19 | 23 | 28 | 31 |
| | 3rd | 3 | 3 | 3 | 4 |

The breakdown of enrolment by major region is presented in Table 3, which also shows the average annual increase in enrolment in absolute numbers. We note the following:

(i) *Developed countries:* The expansion of enrolment at the *first level* declined consistently during the period to become negative after 1970 due to the decrease in the relevant population, both for Northern America and Europe and the USSR. The expansion of *second level* enrolment also declined in Northern America, while in the case of Europe and the USSR it fell from an average of 1.6 million yearly during the period 1960-65 to 1.0 million during 1965-70 and then rose to 1.8 million yearly after 1970. The expansion of *higher education* was the dominant trend in Northern America, and to a lesser extent in Europe and the USSR where the growth picked up again after 1970 following a slow down between 1965 and 1970.

(ii) *Developing countries:* The average yearly absolute increase in enrolment in *first level* in Latin America increased from 1.6 million during 1960-65 to 2.2 million during 1965-70 and then slightly decreased to

2.0 million after 1970. In the case of Africa, the average annual increase during the period 1960-65 registered a relative decrease in the following years and then recovered its growing trend. On the other hand, expansion of enrolment in Asia registered a decreasing path of 5.8 million yearly during 1960-65 to 4.6 million yearly during 1965-70 and 4.7 million during the period 1970-1976.

The average yearly increase in enrolment at the *second level* was very significant in the case of Africa (200,000 yearly during 1960-65, 400,000 during 1965-70 and 700,000 after 1970). It was somewhat stable in Asia (2.2, 2.0 and 2.3 million respectively for the three periods). As for Latin America, the expansion was clear: 400,000 yearly during 1960-65, 600,000 during 1965-70 and 800,000 after 1970.

As regards expansion at the *third level,* the increasing trend in Latin America from 1960 to 1970 (60,000 yearly during 1960-65 and 140,000 during 1965-70) became extremely high (400,000 a year) between 1970 and 1976. The progression was geometric in the case of Africa and arithmetic in Asia.

Because of the differing patterns of expansion at the different levels within each region, the shapes of the educational pyramids were modified. This is for example the case of Northern America with some 81 pupils in second level for each 100 in first level and correspondingly 46 in third level. At the other extreme, Africa has only 20 and 2 respectively. This is illustrated in Figure 2.

2. *Enrolment ratios.* Enrolment ratios relate the number of pupils enrolled in a given age-group or level to a total population of a specific age-group. They permit an assessment of educational participation although it should be noted that the enrolment data refer to registration and not to actual attendance. Gross enrolment ratios in this paper relate enrolment at the regional level in first, second and third levels of education to the population age-groups 6-11, 12-17 and 18-23 respectively, which are the age-groups generally assumed to correspond to the three levels of enrolment. As educational structures differ between countries and since under- and over-aged pupils are enrolled at each level, these ratios only give a crude measure of the coverage at each level. Because of these two factors they may sometimes exceed 100%, i.e. the size of the age-group being used as a yardstick. They do however show the *potential capacity* of the system to enrol the population of a given age-group in the absence of under- and over-aged pupils.

Table 4 presents the set of gross enrolment ratios by level for the period under study. The following summary remarks may be made on the table:

Table 3. *Development of Enrolment by Level of Education and Major Regions, 1960-1976 (in millions)*

| Major Regions | Level | Enrolment | | | | Average absolute annual increase | | |
|---|---|---|---|---|---|---|---|---|
| | | 1960 | 1965 | 1970 | 1976 | 1960-65 | 1965-70 | 1970-76 |
| Northern America | 1st | 29 | 32 | 32 | 28 | .6 | 0 | − .7 |
| | 2nd | 15 | 19 | 22 | 23 | .8 | .6 | .2 |
| | 3rd | 4 | 6 | 9 | 13 | .4 | .6 | .7 |
| | Total | 48 | 57 | 63 | 64 | 1.8 | 1.2 | 0 |
| Europe & USSR | 1st | 79 | 88 | 92 | 83 | 1.8 | .8 | −1.5 |
| | 2nd | 23 | 31 | 36 | 47 | 1.6 | 1.0 | 1.8 |
| | 3rd | 5 | 8 | 10 | 14 | .6 | .4 | .7 |
| | Total | 107 | 127 | 138 | 144 | 4.0 | 2.2 | 1.0 |
| Africa | 1st | 19 | 27 | 34 | 47 | 1.6 | 1.4 | 2.2 |
| | 2nd | 2 | 3 | 5 | 9 | .2 | .4 | .7 |
| | 3rd | .2 | .3 | .5 | 1 | .02 | .04 | .08 |
| | Total | 21.2 | 30.3 | 39.5 | 57 | 1.82 | 1.84 | 2.98 |
| Asia | 1st | 87 | 116 | 139 | 167 | 5.8 | 4.6 | 4.7 |
| | 2nd | 29 | 40 | 50 | 64 | 2.2 | 2.0 | 2.3 |
| | 3rd | 3 | 5 | 7 | 9 | .4 | .4 | .4 |
| | Total | 119 | 161 | 196 | 240 | 8.4 | 7 | 7.4 |
| Latin America | 1st | 28 | 36 | 47 | 59 | 1.6 | 2.2 | 2.0 |
| | 2nd | 3 | 5 | 8 | 13 | .4 | .6 | .8 |
| | 3rd | .6 | .9 | 1.6 | 4 | .06 | .14 | .4 |
| | Total | 31.6 | 41.9 | 56.6 | 76 | 2.06 | 2.94 | 3.2 |

(i) *First level education:* The enrolment in 1960 represented 77% of the population 6-11 years in the world, as compared to 88% in 1976. While the developed countries continued their trend of enrolling their primary school-age population, the developing countries gained 22 percentage points, despite a remarkable population growth. The largest increase in enrolment ratios was in Latin America (31 percentage points), followed by Africa (26 points) and then Asia (14 points).

Table 4. *Gross Enrolment Ratios for First, Second and Third Level of Education in % (both sexes)*

| Major region | First level | | | | Second level | | | | Third level | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1960 | 1965 | 1070 | 1976 | 1960 | 1965 | 1970 | 1976 | 1960 | 1965 | 1970 | 1976 |
| WORLD | 77 | 83 | 85 | 88 | 27 | 32 | 35 | 40 | 5 | 8 | 10 | 12 |
| Developed countries | 111 | 115 | 115 | 112 | 50 | 56 | 50 | 68 | 11 | 17 | 19 | 25 |
| Developing countries | 59 | 70 | 75 | 81 | 15 | 20 | 24 | 28 | 2 | 3 | 4 | 5 |
| N. America | 120 | 120 | 117 | 120 | 72 | 80 | 83 | 83 | 25 | 30 | 38 | 48 |
| Europe & USSR | 114 | 120 | 122 | 118 | 40 | 45 | 49 | 62 | 8 | 14 | 15 | 19 |
| Africa | 45 | 54 | 59 | 71 | 5 | 7 | 10 | 17 | 1 | 1 | 1 | 2 |
| Asia | 60 | 68 | 70 | 74 | 23 | 29 | 31 | 33 | 2 | 4 | 5 | 5 |
| Latin America | 80 | 90 | 102 | 111 | 11 | 16 | 20 | 30 | 3 | 4 | 5 | 11 |

The regional gross enrolment ratios mask large disparities between different developing countries. The distribution in Table 5 allows a more precise view of the progress made by the various countries. Thus, whereas only 13 of the 34 African countries had a rate of 75% and above in 1960, there were 21 countries with comparable rates in 1976. The corresponding number of countries for Asia was 8 and 25 out of 47, and for Latin America 16 and 24, respectively, out of 26.

(ii) *Second level education:* The world enrolment ratio increased from 27% in 1960 to 40% in 1976. At this level, the developed countries gained 18

Table 5. *Distribution of Developing Countries according to their Gross Enrolment ratio, for First Level Education in 1960 and 1976*

| Distribution of gross enrolment ratio 1st level education | Number of African countries | | Number of Asian countries | | Number of Latin American countries | |
|---|---|---|---|---|---|---|
| | 1960 | 1976 | 1960 | 1976 | 1960 | 1976 |
| 50% and below | 12 | 6 | 22 | 13 | 2 | 0 |
| 51 to 74% | 9 | 7 | 17 | 9 | 8 | 2 |
| 75 to 90% | 3 | 8 | 3 | 10 | 2 | 3 |
| Above 90% | 10 | 13 | 5 | 15 | 14 | 21 |
| Total | 34 | 34 | 47 | 47 | 26 | 26 |

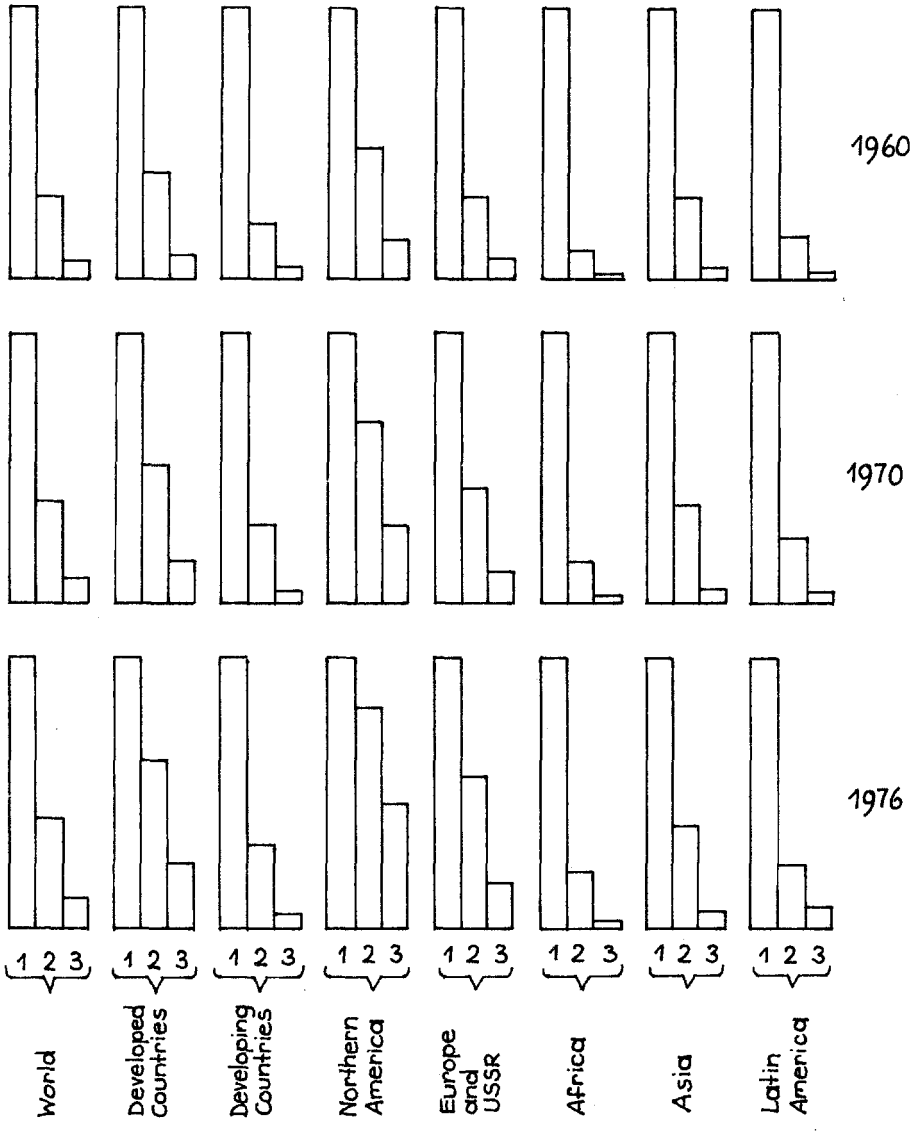



Figure 2. *Educational Pyramid*
*Note 1 = 1st level*
*2 = 2nd level*
*3 = 3rd level*

percentage points, due mainly to Europe and the USSR (22 points) while Northern America increased its already high level by 11 points and attained 83%. In the developing countries, Latin America increased by 19 percentage points followed by Africa (12 points) and Asia (10 points).

(iii) *Third level education:* The gross enrolment ratio at this level increased from 5% to 12% during the 16 years under review for the world as a whole. The largest increase at the regional level was observed in Northern America, i.e. 23 points, followed by Europe and the USSR, 11 points. Latin America increased by 8 points while the increases for Asia and Africa were only 3 and 1 respectively.

3. *Enrolment disparities by sex.* Table 6 presents the enrolment of girls expressed as a percentage of the total, by level, for 1960 and 1976. It can be seen that in the developed countries, participation of girls is close to boys at first and second levels of education. This was already so in 1960. Considerable progress took place at the third level between 1960 and 1976. As for the developing countries, the situation is far from being the same except in the case of Latin America where the pattern is very close at the first and second levels of education and serious progress has been made at the third level. In Africa and Asia although improvements have been made, the gap remains large at all levels and especially so at the second and third levels of education. In 1976, girls represented only one-third of second level enrolment and one-quarter of third level in Africa and the corresponding percentages for Asia were 37% and 32% respectively.

Table 6. *Percentage of girls enrolled by level of education*

| Major Regions | Total | | First level | | Second level | | Third level | |
|---|---|---|---|---|---|---|---|---|
| | 1960 | 1976 | 1960 | 1976 | 1960 | 1976 | 1960 | 1976 |
| WORLD | 43 | 44 | 44 | 45 | 42 | 45 | 33 | 42 |
| Developed countries | 48 | 49 | 49 | 49 | 49 | 50 | 35 | 44 |
| Developing countries | 37 | 42 | 39 | 43 | 29 | 36 | 24 | 33 |
| Africa | 36 | 40 | 36 | 42 | 30 | 35 | 18 | 25 |
| Asia | 37 | 40 | 39 | 42 | 33 | 37 | 23 | 32 |
| Latin America | 48 | 49 | 48 | 49 | 47 | 49 | 30 | 41 |

Perhaps the best way to assess disparity in enrolment by sex consists of subtracting the female enrolment from the male enrolment for the same year and level of education. This is shown in Table 7 below.

Table 7. *Absolute difference in enrolment between boys and girls (in millions)*

| Major Regions | Total | | First level | | Second level | | Third level | |
|---|---|---|---|---|---|---|---|---|
| | 1960 | 1976 | 1960 | 1976 | 1960 | 1976 | 1960 | 1976 |
| WORLD | 44 | 66 | 29 | 37 | 10 | 22 | 5 | 7 |
| Developed countries | 7 | 6 | 3 | 2 | 1 | 0 | 3 | 4 |
| Developing countries | 37 | 60 | 26 | 35 | 9 | 22 | 2 | 3 |
| Africa | 6.2 | 10.8 | 5 | 7 | 1 | 3 | .2 | .8 |
| Asia | 29.6 | 46 | 19 | 27 | 9 | 16 | 1.6 | 3 |
| Latin America | 3.2 | 2.8 | 2 | 1 | 1 | 1 | .2 | .8 |

It can be seen that the gap between boys and girls that was 44 million pupils in 1960 became 66 million in 1976. The trend appears to be towards a decrease in the developed countries and Latin America while it increases considerably in the other developing countries, Asia and Africa.

By level, the developed countries reduced their sex disparities in enrolment at the *first and second levels* of education while an increase was registered of one million at the *third level.* There was a general increase at all levels of education in Africa and Asia while Latin America, reduced the disparity at the *first level,* kept constant the gap at the *second level* and increased it at the *third level.*

4. *Major issues.* Areas of concern closely associated to the intake and transition problems reviewed above are several and we will here mention two which deserve special attention:

(a) Internal efficiency in primary education in developing countries;

(b) Enrolment at the second and third levels of education by major types of education and fields of study.

(i) *Efficiency and drop-out in primary education*

The expansion of enrolment in developing countries is seriously affected by wastage factors, namely drop-out and repetition. The extent to which the flow of pupils through education systems is affected by these two factors appears often to be independent of the level of enrolment. This can be illustrated by the patterns of *drop-out* observed for two cohorts, at the regional level, starting in 1965 and 1970, as shown in Table 8. It should be recalled that as the figures in this table are weighted regional averages the level of drop-out for each region is heavily influenced by the situation in the most populous countries.

Table 8. *Estimated Drop-out Patterns of Developing Countries in First Level Education for Cohorts Starting in 1965 and 1970 (both sexes) (in percentage)*

| Major Region | Year | % of enrolment in Grade 1 dropping-out between: | | | | |
|---|---|---|---|---|---|---|
| | | Grades 1 and 2 | Grades 2 and 3 | Grades 3 and 4 | Grades 4 and 5 | Total drop-out Grades 1 to 5 |
| Developing | 1965 | 33 | 9 | 8 | 9 | 59 |
| countries | 1970 | 30 | 9 | 7 | 9 | 55 |
| Africa | 1965 | 22 | 7 | 10 | 9 | 48 |
| | 1970 | 17 | 5 | 6 | 8 | 36 |
| Asia (South) | 1965 | 33 | 9 | 9 | 9 | 60 |
| | 1970 | 31 | 9 | 8 | 11 | 59 |
| Latin America | 1965 | 42 | 10 | 4 | 11 | 67 |
| | 1970 | 36 | 10 | 7 | 6 | 59 |

It can be seen that (a) the overall percentage drop-out is generally high but decreases between the two cohorts; (b) drop-out is much higher in South Asia and Latin America (59%) in 1976 than in Africa (36%); (c) the drop-out is very high between the first and second grades, especially in Latin America and South Asia.

As regards repetition, recent estimates show that around 1976, the proportion of total enrolment at the *first level* of education constituted by repeaters was around 14% in Africa and Asia, and 12% in Latin America.

(ii) *Enrolment by type and field of study*

Enrolment at the *second level of education* in the world in 1976 was distributed as follows by type: General, 84.6%; Teacher Training, 1.4% and Other (mainly Vocational and Technical Education), 14.0%. This distribution which remained rather stable over the period 1960-1976, is different for developed and developing countries (81%, 1% and 18% and 88%, 2% and 10% respectively), the latter countries thus having a heavier representation at the General type of education.

As regards the *third level of education,* Table 9 shows the distribution of enrolment by field of study in 1975 grouping into three major *categories*; (a) Scientific and technological fields (Natural Sciences, Engineering, Medical Sciences and Agriculture); (b) Other fields (Humanities, Education, Fine Arts, Law and Social Sciences) and (c) Not specified.

It is interesting to note the relatively small differences between the different regions as regards the scientific fields' share of enrolment at this level.

Table 9. *Enrolment at the Third Level of Education by Field of Study, 1975 (in percentage)*

| | Scientific and technological fields | Other fields | Not specified |
|---|---|---|---|
| World | 40.3 | 55.8 | 3.9 |
| Developed countries | 41.2 | 55.4 | 3.3 |
| Developing countries | 38.2 | 56.6 | 5.3 |
| Africa | 37.8 | 61.2 | 1.0 |
| Asia | 35.0 | 62.4 | 2.6 |
| Latin America | 38.7 | 50.7 | 10.6 |

*Teaching Staff*

The number of teaching staff in the world grew during the period at a faster rate than enrolment. Thus, the total of 12.7 million teachers at all levels of education virtually doubled to reach 25.3 million. Of the net increase, 5.4 million joined the first level, 5.3 million the second level and 1.9 million the third level. Trends and multipliers as shown in Table 10 indicate a higher relative growth in developing than in developed countries, at all levels of education. In spite of this, the former countries had in 1976 more than 58% of the total enrolment with only 46% of the world's teaching staff.

Table 11 below presents trends in *pupil-teacher ratios* for the three levels of education. These ratios show at the *first level of education:* (a) Higher numbers of pupils per teacher in developing than in developed countries. There are also large differences between the ratios for each of the developing regions. It is striking to note that in 1976 Africa and, to a lesser extent, Asia had almost twice as many pupils per teacher as in the developed countries; (b) A general decrease in the ratios over the period considered in all regions except Africa and Asia where values remained practically stable. At the *second level of education,* the difference between the ratios is smaller between developed and developing countries with the latter countries generally having higher values. Furthermore, while the developed countries tend to reduce their ratios, Africa increased them rather considerably while

Table 10. *Teaching Staff by Level of Education (in millions) 1960 and 1976*

| Major region | First level | | Second level | | Third level | | Total | | Multiplier by level | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1960 | 1976 | 1960 | 1976 | 1960 | 1976 | 1960 | 1976 | 1st | 2nd | 3rd | Total |
| WORLD | 8.2 | 13.6 | 3.6 | 8.9 | 0.9 | 2.8 | 12.7 | 25.3 | 1.7 | 2.5 | 3.1 | 2.0 |
| Developed countries | 4.9 | 6.5 | 2.4 | 5.1 | 0.7 | 2.0 | 8.0 | 13.6 | 1.3 | 2.1 | 2.9 | 1.7 |
| Developing countries | 3.3 | 7.1 | 1.2 | 3.8 | 0.2 | 0.8 | 4.7 | 11.7 | 2.2 | 3.2 | 4.0 | 2.5 |
| N. America | 1.1 | 1.4 | 0.6 | 1.3 | 0.3 | 0.7 | 2.0 | 3.4 | 1.3 | 2.2 | 2.3 | 1.7 |
| Europe & USSR | 3.3 | 4.0 | 1.4 | 3.2 | 0.3 | 1.0 | 5.0 | 8.2 | 1.2 | 2.3 | 3.3 | 1.6 |
| Africa | 0.5 | 1.2 | 0.1 | 0.4 | 0.01 | 0.06 | 0.6 | 1.7 | 2.4 | 4.0 | 6.0 | 2.8 |
| Asia | 2.5 | 4.7 | 1.2 | 3.0 | 0.2 | 0.6 | 3.9 | 8.3 | 1.9 | 2.5 | 3.0 | 2.1 |
| Latin America | 0.8 | 2.1 | 0.3 | 0.9 | 0.07 | 0.4 | 1.2 | 3.4 | 2.6 | 3.0 | 5.7 | 2.8 |

Asia experienced a slight decrease and Latin America, with very low ratios at the beginning of the period registered a gradual increase.

As for *third level education* perhaps the most remarkable feature is the similarity of the ratios for all regions considered, the only exception being Latin America where the rates were systematically lower than for the other regions, which was also the case at second level.

To sum up, the major difference in provision of teaching staff between developed and developing countries is found at the first level which obviously influenced the structure of unit cost by level of education.

### *Public Expenditure on Education*

Public expenditure on education in the world at current market prices rose from 51 billion U.S. dollars in 1960 to 366 billion in 1976. The share of the developed countries in this total which in 1960 was 91% became 87% in 1976 whereas as indicated in Table 12 their total public expenditure rose from 46.7 billion to 318.1 billion U.S. dollars.

Considering the difficulty in comparing the data at this level of aggregation (mainly because of the difficulty of conversion of national currency into U.S. dollars) it is important to warn against conclusions derived from such time series, especially during a period heavily affected by inflation. For this

Table 11. *Pupil-teacher Patios by Level of Education*

| Major Regions | First level | | | | Second level | | | | Third level | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1960 | 1965 | 1970 | 1976 | 1960 | 1965 | 1970 | 1976 | 1960 | 1965 | 1970 | 1976 |
| WORLD | 30 | 30 | 30 | 28 | 20 | 19 | 18 | 18 | 14 | 14 | 14 | 15 |
| Developed countries | 26 | 24 | 23 | 20 | 20 | 18 | 16 | 16 | 14 | 14 | 14 | 15 |
| Developing countries | 36 | 38 | 36 | 35 | 19 | 20 | 20 | 20 | 16 | 16 | 16 | 14 |
| N. America | 28 | 27 | 24 | 20 | 25 | 21 | 19 | 18 | 13 | 14 | 15 | 18 |
| Europe & USSR | 24 | 23 | 22 | 20 | 17 | 16 | 15 | 15 | 15 | 15 | 13 | 13 |
| Africa | 39 | 40 | 41 | 39 | 17 | 20 | 22 | 24 | 16 | 15 | 14 | 16 |
| Asia | 36 | 38 | 36 | 35 | 23 | 23 | 22 | 21 | 16 | 17 | 17 | 15 |
| Latin America | 34 | 32 | 32 | 28 | 12 | 13 | 14 | 14 | 8 | 8 | 10 | 11 |

Table 12. *Public Expenditure on Education, 1960-1976 (in billions, of current U.S. dollars) and as a percentage of Gross National Product*

| Major region | Public expenditure on education (billions U.S. $) | | | | as % of G.N.P. | | | |
|---|---|---|---|---|---|---|---|---|
| | 1960 | 1965 | 1970 | 1976 | 1960 | 1965 | 1970 | 1976 |
| WORLD | 51.2 | 96.6 | 161.6 | 366.2 | 3.8 | 4.9 | 5.4 | 5.8 |
| Developed countries | 46.7 | 88.2 | 146.5 | 318.1 | 4.0 | 5.2 | 5.6 | 6.0 |
| Developing countries | 4.5 | 8.4 | 15.1 | 48.1 | 2.3 | 3.2 | 4.0 | 4.8 |
| N. America | 21.0 | 40.0 | 71.5 | 125.7 | 3.8 | 5.4 | 6.6 | 6.2 |
| Europe & USSR | 23.3 | 43.1 | 64.7 | 153.7 | 4.2 | 5.1 | 5.1 | 5.9 |
| Africa | 0.8 | 1.3 | 2.3 | 7.7 | 2.7 | 3.4 | 4.2 | 5.0 |
| Asia | 3.8 | 7.7 | 15.7 | 58.1 | 2.9 | 3.7 | 4.3 | 5.6 |
| Latin America | 1.7 | 3.4 | 5.3 | 13.7 | 2.2 | 3.1 | 3.3 | 3.4 |

reason, the table also presents the percentage that public expenditure on education represents when compared to the Gross National Product. A further limitation of these indicators lies in the fact that expenditure is limited to public sources (as private expenditure on education is insufficiently known) while our analysis on enrolment is comprehensive.

We will consider successively two aspects of public expenditure on

education as percentage of G.N.P.: (a) levels in 1960 and (b) increases during 1960-1976 and levels in 1976.

*In 1960:* Public expenditure on education in the developed countries represented 4% of their G.N.P. (Europe and the USSR had a percentage of 4.2% while Northern America registered 3.8%). The developing countries with an overall percentage of 2.3 ranged from 2.9% in Asia to 2.2% in Latin America.

*In 1976:* The developed countries raised their expenditure on education as a percentage of their G.N.P. by 2 percentage points to reach 6% (6.2% in the case of Northern America and 5.9% in the case of Europe and the USSR). The developing countries raised their percentage to 4.8%, i.e. an increase of 2.5 percentage points during the period, with Asia increasing to 5.6% in 1976, Africa to 5% and Latin America to 3.4% as can be seen in Figure 3. The figure also indicates the very fast increase of this percentage for the developed countries in the period 1960-65 followed by a moderate increase in the two subsequent periods. The developing countries as a whole registered a steady rise in the percentage due mainly to Asia, while Africa maintained a stable progress. As for Latin America, the trend after 1965 was towards stabilisation of the percentage.

Two interrelated major features as regards public expenditure on education are worth highlighting. Firstly, during the period under review, most governments, particularly in developing countries, spent an increasing part of their GNP on education. This rise was especially important in Asia. Secondly, in 1976, the *average overall public expenditure per pupil,* as well as the percentage that this average constituted of GNP per capita differed considerably between regions, as can be seen below:

| | (1) Per pupil expenditure | (2) GNP per capita | $\frac{\text{Column 1}}{\text{Column 2}}$ (%) |
|---|---|---|---|
| | (US $) | (US $) | (US $) |
| Northern America | 1,790 | 8,533 | 21 |
| Europe and USSR | 930 | 3,542 | 26 |
| Asia | 255 | 764 | 33 |
| Latin America | 172 | 1,215 | 14 |
| Africa | 149 | 401 | 37 |

Thus, in 1976, overall expenditure per pupil on education represented 21% of the GNP per capita in Northern America, 26% in Europe and the USSR, 33% in Asia, 14% in Latin America and 37% in Africa. One factor which

goes towards explaining the relatively low percentage in Latin America might be that private expenditure on education is relatively high in several countries in this region. Clearly, the very high level of *per pupil expenditure* as compared with *per capita income* in a large number of countries in Africa and Asia may place serious constraints on the future expansion of education in these regions, particularly since the relatively small monetary sector of their economies does not offer the same possibilities as in industrialized nations for raising public income through taxes. This applies also to several Latin American countries.

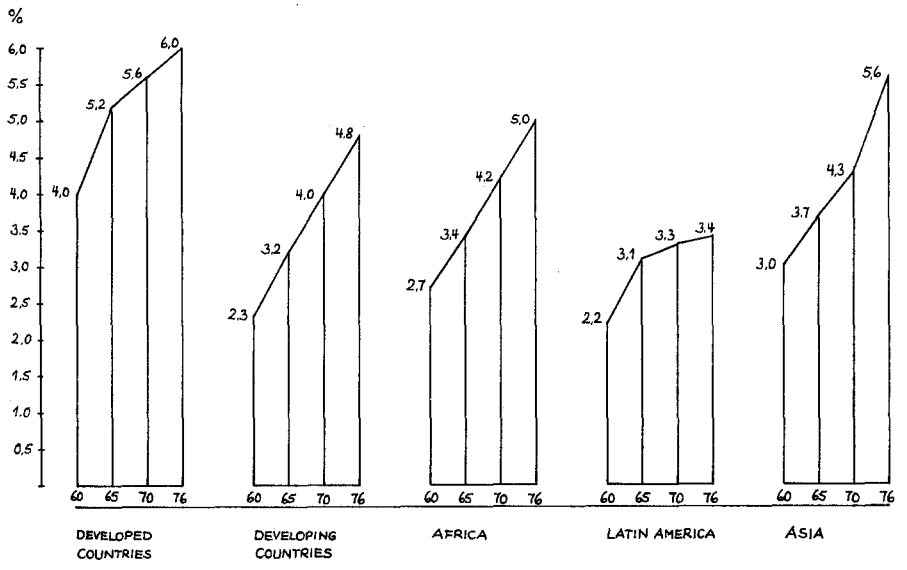



Figure 3. *Public expenditure on education as percentage of G.N.P.*

## *Concluding Remarks*

The primary purpose of this summary statistical review has been to present major facts and features in the development of education since 1960.

A major issue indicated above was the demographic factor. Population changes will in the future, as was the case in the past, continue to affect the ability of countries to respond to the growing demand for education in a world in constant process of modernisation. In this connection, educational planners will need to develop alternative approaches and methodologies in order to better evaluate present realities and efficient strategies for the

future. This is definitely true for both the developed and the developing countries.

The significant expansion in enrolment implied a rise of 77% of total enrolment at the first, second and third levels of education between 1960 and 1976. The rise was of 32% in the developed and of 133% in the developing countries. However, disparities between developed and developing countries continued to be marked. Thus, *developed countries,* besides accomplishing practically universal primary and secondary education, experienced an exceptional boom at the third level. This is particularly so in Northern America.

As for *developing countries,* despite a sharp increase of enrolment at the first level of education, late entrants and repeaters inflate their enrolment ratios, partly masking the increasing number of out-of-school population in a large number of countries. Furthermore, drop-out and repetition continue to be serious bottlenecks to the efficient utilization of available resources. Expansion has been noteworthy at the second level in Asia and at the third level in Latin America. They, however, remain proportionately moderate as is shown by their school pyramids.

Although the percentage of total enrolment constituted by girls increased at all levels in all regions, the gap between the two sexes measured in absolute terms increased in several cases, namely at the third level in the developed countries, at all levels of education in Africa and Asia but only at the third level of education in Latin America.

In 1976, the developing countries had a higher percentage of their *second level enrolment* in General Education and a lower part in vocational and technical education than did the developed countries. This is contrary to the situation in *higher education* where the share of scientific and technological fields in total enrolment did not differ much between regions.

The number of *teaching staff* in the world grew at a faster rate than enrolment, the rate of growth being higher in developing than in developed countries, at all levels of education. Developing countries however, had in 1976 more than 58% of the total enrolment with only 46% of the world's teaching staff. The major difference in provision of teaching staff between developed and developing countries is found at the first level where there are higher numbers of pupils per teacher in the latter region.

*Public expenditure on education* in the world, at current market prices, grew from 51 billion U.S. dollars in 1960 to 366 billion in 1976. There is no doubt that education remained one of the prioritary sectors for public concern, since the average annual growth rate of educational expenditure for the period was higher than that of the Gross National Product (13.1% as

compared to 10.1%). This was the case for the developed countries (12.7% versus 9.9%) and even more so far the developing countries (16% versus 10.9%). As a result in 1976, public expenditure on education represented 5.8% of the Gross National Product for the world, compared with 3.8% in 1960. The figure for the developed countries was 6% in 1976 and 4% in 1960 compared to 4.8% and 2.3% respectively in the developing countries. Furthermore, the relatively high level of per pupil public expenditure as compared with per capita income in most African and Asian countries as well as in several Latin American countries, appears to be a serious constraint to potential expansion of education in these regions.

In conclusion, social demand – heavily influenced in the developing countries by population pressures – will continue to place emphasis on educational needs. Extension of basic education and literacy to disadvantaged groups and areas will no doubt become a priority as will the reconsideration of educational content to match development needs. Out-of-school and adult education will have to be taken more into account without, however, placing too many expectations upon their capacity to provide learning at lower costs than formal education.

Finally it should be further realized that sound and pragmatic educational planning is only possible when based upon a solid, consistent and comprehensive statistical data base.

*This article expresses the views of the author and not necessarily those of Unesco or of the Editorial Board.*